SE8215

MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)

UG Programme (Honors) Degree March 2025 Regular Examinations

Second Semester

242CCRUBC103 Operating Systems

Time: **2 Hours** Maximum Marks: **40**

Answer one question from each bunch

BUNCH I

I.A.1 Mr. Akhil gives a command to execute a program sum.exe. Illustrate the state changes possible to this program during its entire execution time. (CO2:5 Marks)

I.A.2 Assume that the following processes have to be executed with one processor, with the processes arriving in the order listed below.

Process ID	Burst Time	Arrival Time
0	25	0
1	15	10
2	10	10
3	20	15

Suppose a system uses RR scheduling with a quantum of 5. Create a Gantt chart illustrating the execution of these processes.

i. What is the turnaround time of processes P2 and P3?

ii. What is the average waiting time of the processes? (CO2:5 Marks)

OR

I.B Differentiate between the following:

1. Short-term Scheduler and Medium-term Scheduler
2. Preemptive Scheduling and Non-preemptive Scheduling

3. Job Queue and Ready Queue
4. Process and Thread (CO2:10 Marks)

BUNCH II

II.A.1 Consider a system with 12 magnetic tape drives and three processes: P0, P1, and P2. Process P0 requires 10 tape drives, process P1 may need as many as 4 tape drives, and process P2 may need up to 9 tape drives. Suppose that, at time T0, we have:

Process	Maximum Need	Current Allocation
P0	10	5
P1	4	2
P2	9	2

Currently, there are 3 tape drives available in the system. Using the Banker's Algorithm, determine whether the system is in a safe state. If a new request for 1 tape drive by P2 arrives, can it be granted safely? Explain your answer using the steps of the deadlock avoidance algorithm. (CO3:5 Marks)

II.A.2 Demonstrate how the use of Peterson's Algorithm ensures mutual exclusion in a two-process system. (CO3:5 Marks)

OR

II.B.1 A hospital management system allows multiple processes to access limited MRI machines and printers. One process is printing patient reports, while another is scanning MRI results. Each needs both the printer and scanner, but they each hold one and wait for the other. What condition of deadlock is illustrated here? How can this situation result in a deadlock? (CO3:5 Marks)

II.B.2 A process P1 is executing and currently holds access to a file resource. During its execution, it attempts to request access to a printer. However, the operating system follows a deadlock prevention strategy in which it enforces a policy that denies any new resource request if the process is already holding one or more resources. (CO3:5 Marks)

II.B.(a). Explain how this policy helps in preventing deadlocks, specifically referring to the Hold and Wait condition.

II.B.(b). Discuss two major disadvantages of using this strategy in real-world systems, for system performance and resource utilization.

BUNCH III

III.A Consider the Pages referenced by the CPU in the order 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, for a memory with 3 frames.
Use the following page replacement algorithms and find the no of page faults in each case:

a) FIFO
b) LRU
c) Optimal (CO4:10 Marks)

OR

III.B Consider the following page reference string: 1, 2, 3, 4, 5, 3,4,1,6,7,8,7, 8, 9, 7, 8. How many page faults would occur for the following replacement algorithms, assuming four frames? All frames are initially empty.

III.B.1 LRU replacement (CO4:3 Marks)

III.B.2 FIFO replacement (CO4:3 Marks)

III.B.3 Optimal replacement. (CO4:4 Marks)

BUNCH IV

IV.A.1 The file organization module is essential for the efficient, secure, and reliable management of files within a file system. It ensures that files are stored and accessed in an optimal way while maintaining the integrity and security of the data. With a neat diagram, show the File system structure. (CO5:5 Marks)

IV.A.2 Explain the internal structure of a magnetic disk with the help of a neat diagram. Describe the key physical and logical components, such as platters, tracks, sectors, and cylinders. Also, elaborate on how data is stored, accessed, and organized on a magnetic disk. Include an

explanation of how the read/write head functions in coordination with other parts of the disk during data retrieval and storage operations. (CO5:5 Marks)

OR

IV.B.1 The disk queue contains the following requests in the order they arrive: 98, 183, 37, 122, 14, 124, 65, 67. The initial position of the disk head is 53. Using the FCFS disk scheduling algorithm, determine the total head movement required to service all requests. Also, list the order in which the requests will be serviced. (CO5:5 Marks)

IV.B.(a). Discuss the need for disk scheduling algorithms. How do they contribute to the efficiency of disk I/O operations?

IV.B.2 Assume that you are a system administrator optimizing disk I/O performance for a high-load enterprise server. The server handles a large number of simultaneous disk access requests per second from multiple applications and users. Currently, the server uses the LOOK disk scheduling algorithm. The request queue contains the following disk cylinder numbers: 95, 180, 34, 119, 11, 123, 62, 64, 150, 38, with the initial head position at 100. During peak load, performance inefficiencies are observed. Would switching to the C-LOOK scheduling algorithm help reduce these inefficiencies? Justify your answer based on how C-LOOK handles such scenarios compared to LOOK. (CO5:5 Marks)

CO1	Describe the structure, types, and services of operating systems.
CO2	Analyze the performance of various process Scheduling Algorithms.
CO3	Appraise various techniques for process synchronization and deadlock handling.
CO4	Analyze the methods employed for memory management in computer systems
CO5	Analyse various disk scheduling and understand file system concepts.